

LAB EXPERIMENT RECORD

Department	AIML
Course Code / Course Name	DSA0601 / Data Handling and Visualization
Experiment	Data Visualization using R
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Date	29 September 2026
Day	Tuesday

Set 3: Employee Performance Evaluation

DSA0601 • DATA HANDLING AND VISUALIZATION

Dataset: Employee Performance

Employee ID	Department	Years of Service	Performance Score
1	Sales	5	85
2	HR	3	92
3	Marketing	7	78
4	Sales	4	90
5	HR	2	76

1. Create a line chart to visualize the performance trend of employees over time. Include a legend and labels.
2. Generate a bar chart showing the distribution of employees across different departments. Label the chart elements.
3. Build a scatter plot to analyze the correlation between years of service and performance scores. Explain any insights.
4. Develop a Tableau dashboard with the line chart and bar chart for interactive exploration.

Set 3 Assignment: Employee Performance Evaluation

Dataset: Employee Performance

Employee ID	Department	Years of Service	Performance Score
1	Sales	5	85
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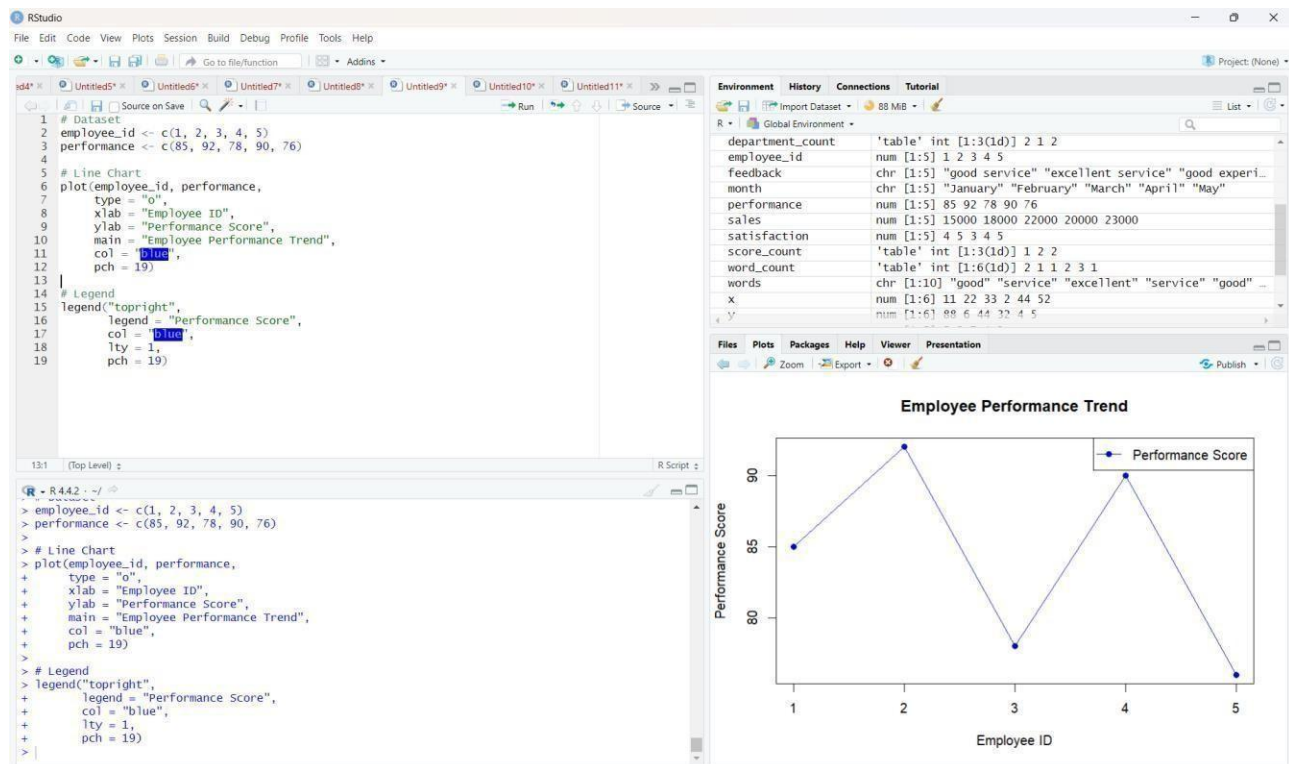
PROGRAM 1: Employee Performance Line Chart

Create a line chart to visualize the performance trend of employees. Include a legend and labels.

```
# Dataset
employee_id <- c(1, 2, 3, 4, 5)
performance <- c(85, 92, 78, 90, 76)

# Line Chart
plot(employee_id,
     performance, type = "o", xlab =
     "Employee ID", ylab = "Performance
     Score", main =
     "Employee Performance Trend", col =
     "blue", pch = 19)

# Legend
legend("topright", legend =
     "Performance Score", col = "blue",
     lty = 1, pch = 19)
```



Output 1: Employee Performance Trend

Observation: Employee 2 has the highest performance score of 92, while Employee 5 has the lowest score of 76.

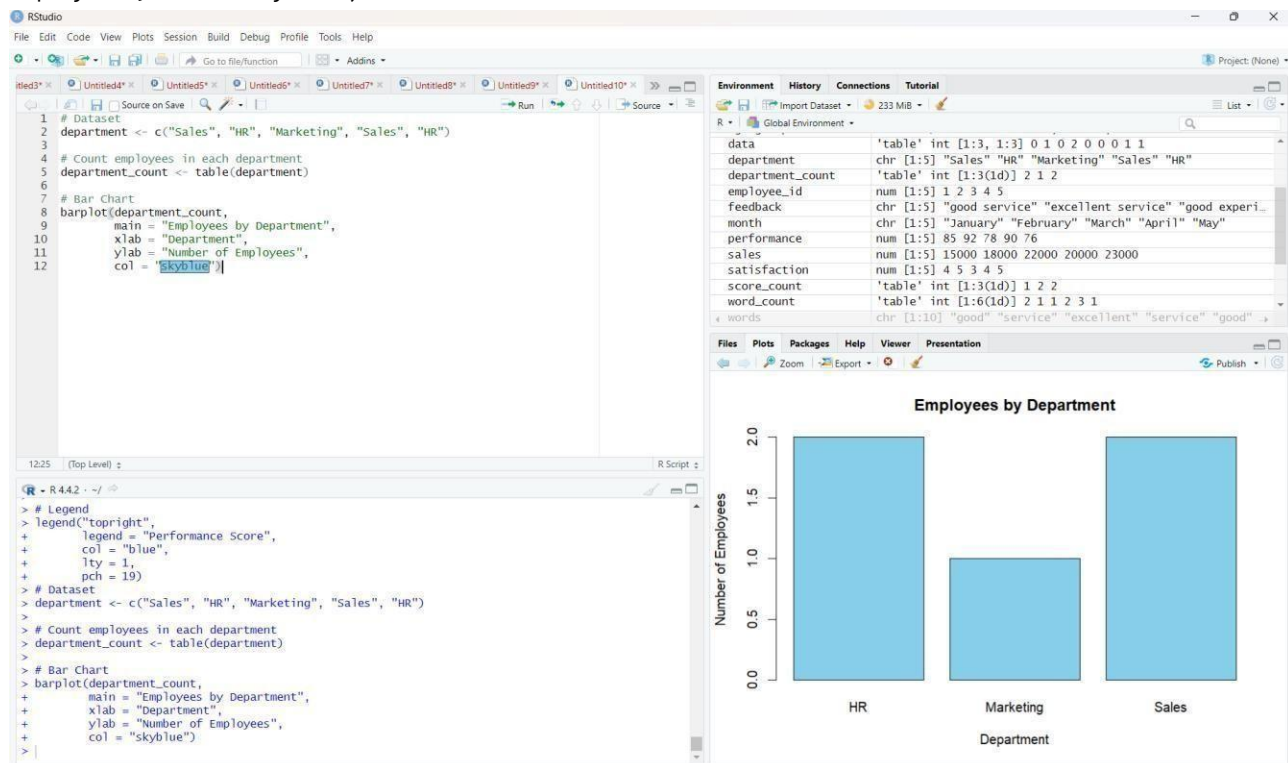
PROGRAM 2: Department Bar Chart

Generate a bar chart showing the distribution of employees across different departments.

```
# Dataset
department <- c("Sales", "HR", "Marketing",
               "Sales", "HR")
```

```
# Count employees in each department
department_count <- table(department)

# Bar Chart barplot(department_count,
main = "Employees by Department", xlab =
"Department", ylab = "Number of
Employees", col = "skyblue")
```



Output 2: Employees by Department

Observation: HR and Sales each have 2 employees, while Marketing has 1 employee.

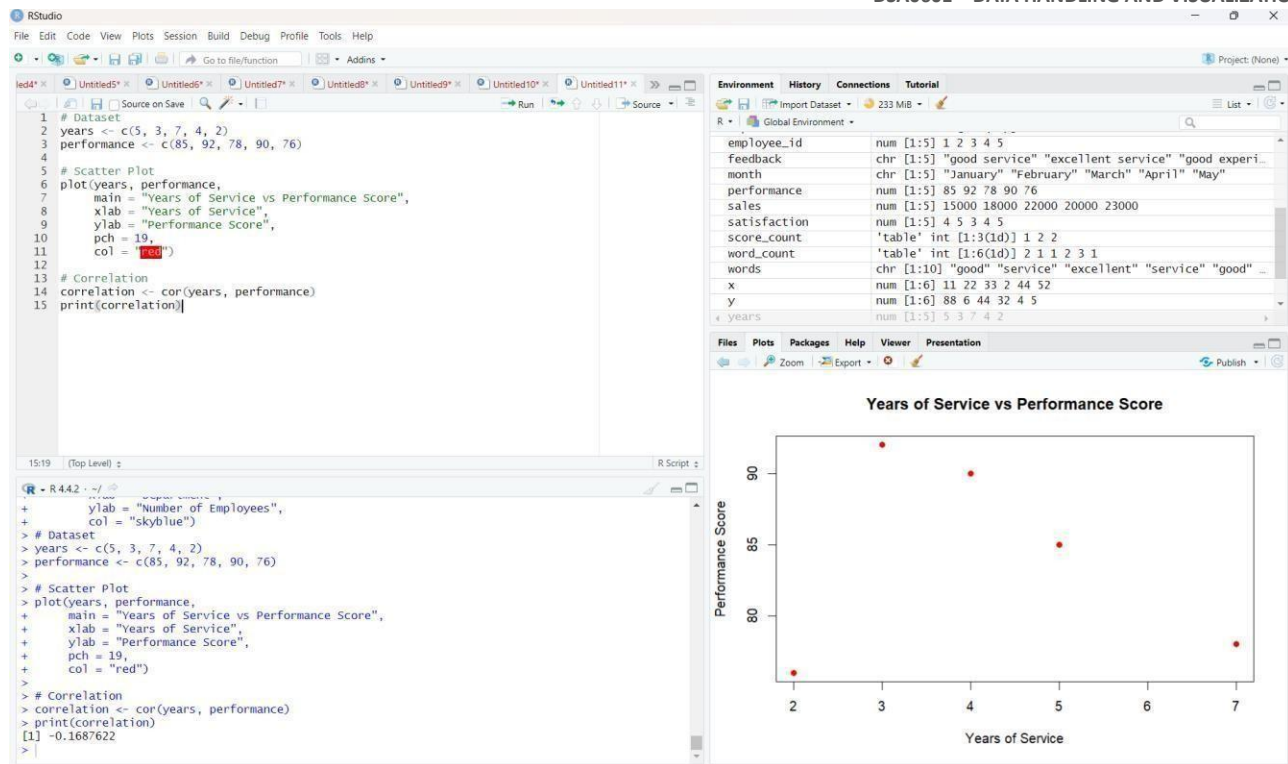
PROGRAM 3: Scatter Plot and Correlation

Build a scatter plot to analyze the correlation between years of service and performance scores.

```
# Dataset years <- c(5, 3, 7, 4, 2)
performance
<- c(85, 92, 78, 90, 76)

# Scatter Plot plot(years, performance, main = "Years
of Service vs Performance Score", xlab = "Years of
Service", ylab = "Performance Score", pch = 19, col
= "red")

# Correlation
correlation <- cor(years, performance)
print(correlation)
```

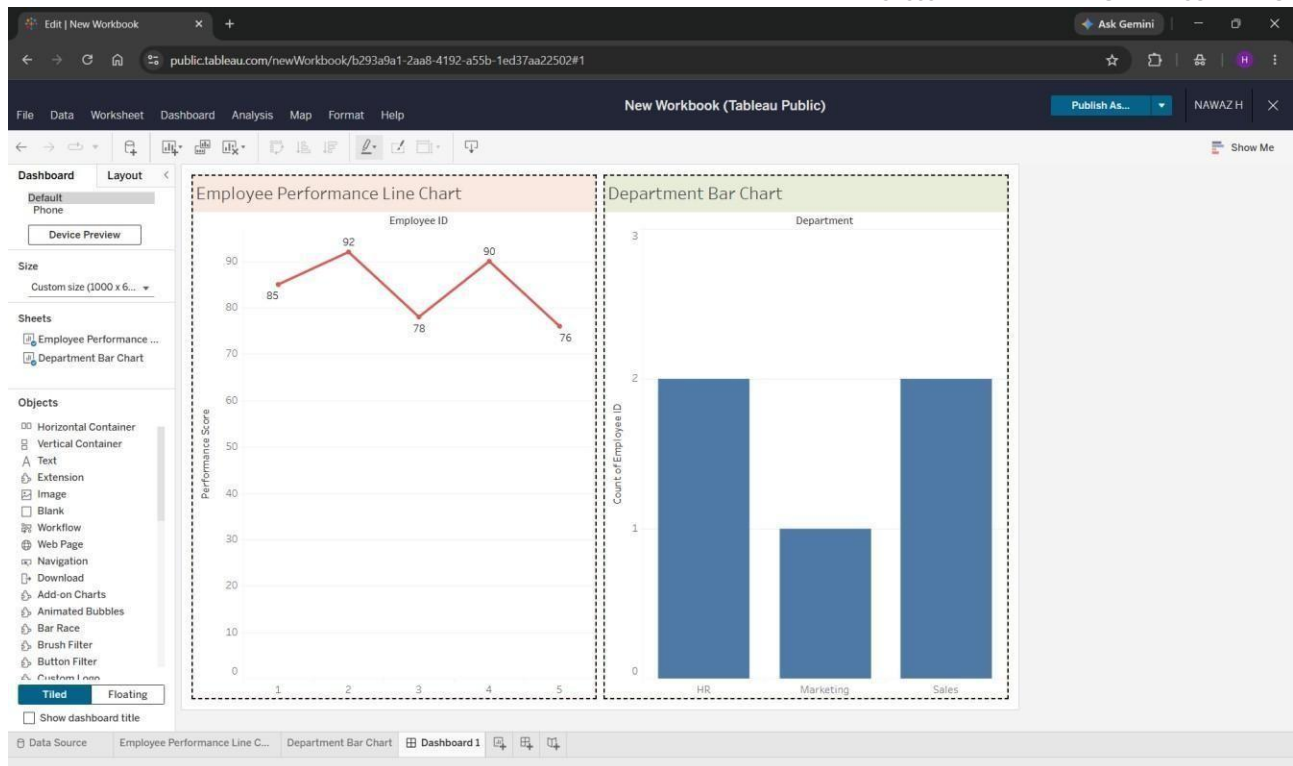


Output 3: Years of Service vs Performance Score

Observation: The calculated correlation is approximately -0.1688, indicating a weak negative linear relationship between years of service and performance score for this small dataset.

PROGRAM 4: Tableau Dashboard

Develop a Tableau dashboard combining the employee performance line chart and department bar chart for interactive exploration.



Output 4: Employee Performance Tableau Dashboard

Observation: The dashboard combines the line chart and department bar chart in one view, allowing the two visualizations to be explored together.

Procedure

1. Create the Employee Performance dataset with Employee ID, Department, Years of Service, and Performance Score.
2. Use `R plot()` to create the employee performance line chart with labels and a legend.
3. Use `table()` and `barplot()` to count employees in each department and display the distribution.
4. Use `plot()` to create the scatter plot and `cor()` to calculate the correlation between years of service and performance score.
5. Open Tableau, create the line chart and department bar chart, and combine them in a dashboard.
6. Add suitable titles, borders, labels, and formatting to make the dashboard easy to read.

Result

The Employee Performance dataset was successfully visualized using a line chart, department bar chart, and scatter plot. The correlation was calculated in R, and a Tableau dashboard was created by combining the line and bar charts.

Key Functions Used

Function / Tool	Purpose
<code>plot()</code>	Creates line and scatter plots in R.
<code>legend()</code>	Adds a legend to the line chart.
<code>table()</code>	Counts employees in each department.
<code>barplot()</code>	Creates the department bar chart.
<code>cor()</code>	Calculates the correlation coefficient.

Tableau Dashboard	Combines charts for interactive visual exploration.
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